

- **Course / Section:** ICS344 F11
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- **Your DVSA Website URL:** <http://dvsa-website-test-053108159524-us-east-1.s3-website.us-east-1.amazonaws.com>
- **AWS Region:** us-east-1
- **Date:** 27/4/2026
- **Vulnerability Title(s):** Over-Privileged Function (Lesson 7)

Part 1) Goal and Vulnerability Summary:

This lesson demonstrates the Over-Privileged Lambda Execution Role vulnerability, where the Lambda function `DVSA-SEND-RECEIPT-EMAIL` is assigned an IAM role with permissions far beyond what it requires.

The **main affected component** is the IAM execution role attached to the Lambda function.

Security impact: an attacker who gains access to the function's temporary credentials can perform unauthorized actions such as reading DynamoDB tables, listing S3 buckets, or sending arbitrary SES emails.

The **weakness** arises from broad wildcard permissions (`s3:*`, `dynamodb:*`, `ses:*`) that violate the principle of least privilege.

Part 2) Why This Works / Root Cause:

The vulnerability exists because the Lambda execution role is configured with excessive permissions that are not required for its intended purpose. Instead of granting only the minimal SES and CloudWatch Logs permissions, the role includes full administrative access to multiple AWS services. The root cause is a failure to enforce least-privilege IAM policies.

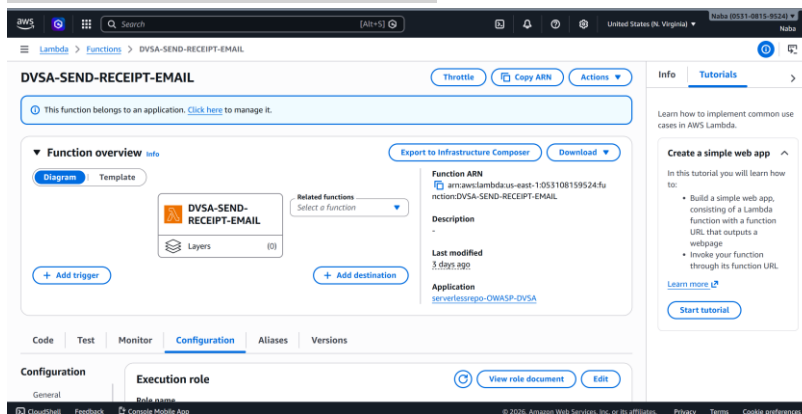
Part 3) Environment and Setup:

- DVSA URL: <http://dvsa-website-test-053108159524-us-east-1.s3-website.us-east-1.amazonaws.com>
- AWS Services Involved:
 - Lambda Function: DVSA-SEND-RECEIPT-EMAIL
 - IAM Execution Role: Attached to the above Lambda
 - CloudWatch Log Group: /aws/lambda/DVSA-SEND-RECEIPT-EMAIL
 - S3 Bucket: dvsa-receipts-bucket-053108159524-us-east-1 (used to trigger the Lambda)
- Tools Used:
 - AWS Console (IAM, Lambda, CloudWatch, CloudTrail)
 - AWS CLI commands were executed inside Ubuntu (WSL) terminal on my local machine.
 - Browser: used to upload the receipt file and trigger the Lambda

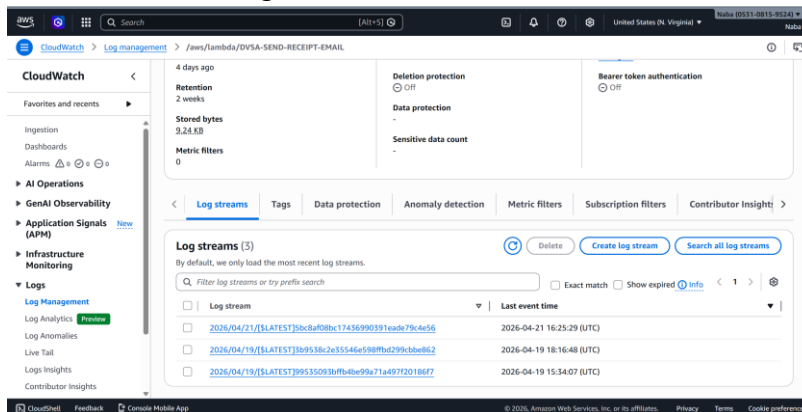
- User Context:
 - Normal DVSA user to trigger the Lambda
 - Attacker profile using temporary credentials printed in CloudWatch logs

Part 4) Reproduction Steps:

1. Navigate to the DVSA website and upload any file to trigger the DVSA-SEND-RECEIPT-EMAIL Lambda function.



2. Go to AWS Console → Lambda → DVSA-SEND-RECEIPT-EMAIL → Monitor → View logs in CloudWatch.



3. Open the most recent Log Stream.
4. Locate the printed Python dictionary containing environment variables.
5. Observe that the log includes sensitive fields such as:
 - a. AWS_ACCESS_KEY_ID
 - b. AWS_SECRET_ACCESS_KEY
 - c. AWS_SESSION_TOKEN
6. Using the AWS CLI, attempt to perform unauthorized actions:


```
aws dynamodb scan --table-name DVSA-ORDERS-DB --region us-east-1
```

7. The command succeeds, proving that the Lambda role has excessive permissions.

Part 5) Evidence and Proof:

1. CloudWatch Log Evidence

The first screenshot shows a list of log events for the function `DVSA-SEND-RECEIPT-EMAIL`. The messages include:

- INIT_START Runtime Version: python3.8.v138 Runtime Version ARN: arn:aws:lambda:us-east-1:runtime:091200f09127049f09026019f9557034b07688f239586840d15bc...
- START RequestId: 5f4f5257-24c5-454f-9544-3842b67f4bcb Version: \$LATEST
- ['AWS_LAMBDA_FUNCTION_VERSION': '\$LATEST', 'AWS_SESSION_TOKEN': 'IQo3b3p221ukVjEHEACVLUWnc3QHS3HHEUCID3s132+4TFfE7H5Rmuc0CyLlNA/sizfgz27JN8wAIEA+V...
- LAMBDA_WARNING: Unhandled exception. The most likely cause is an issue in the function code. However, in rare cases, a Lambda runtime update can cause unex...
- [ERROR] IndexError: list index out of range Traceback (most recent call last): File "/var/task/send_receipt_email.py", line 27, in lambda_handler use...
- END RequestId: 5f4f5257-24c5-454f-9544-3842b67f4bcb
- REPORT RequestId: 5f4f5257-24c5-454f-9544-3842b67f4bcb Duration: 6.61 ms Billed Duration: 385 ms Memory Size: 128 MB Max Memory Used: 56 MB Init Duration: ...
- START RequestId: 5f4f5257-24c5-454f-9544-3842b67f4bcb Version: \$LATEST
- LAMBDA_WARNING: Unhandled exception. The most likely cause is an issue in the function code. However, in rare cases, a Lambda runtime update can cause unex...
- [ERROR] IndexError: list index out of range Traceback (most recent call last): File "/var/task/send_receipt_email.py", line 27, in lambda_handler use...
- ['AWS_LAMBDA_FUNCTION_VERSION': '\$LATEST', 'AWS_SESSION_TOKEN': 'IQo3b3p221ukVjEHEACVLUWnc3QHS3HHEUCID3s132+4TFfE7H5Rmuc0CyLlNA/sizfgz27JN8wAIEA+V...
- END RequestId: 5f4f5257-24c5-454f-9544-3842b67f4bcb

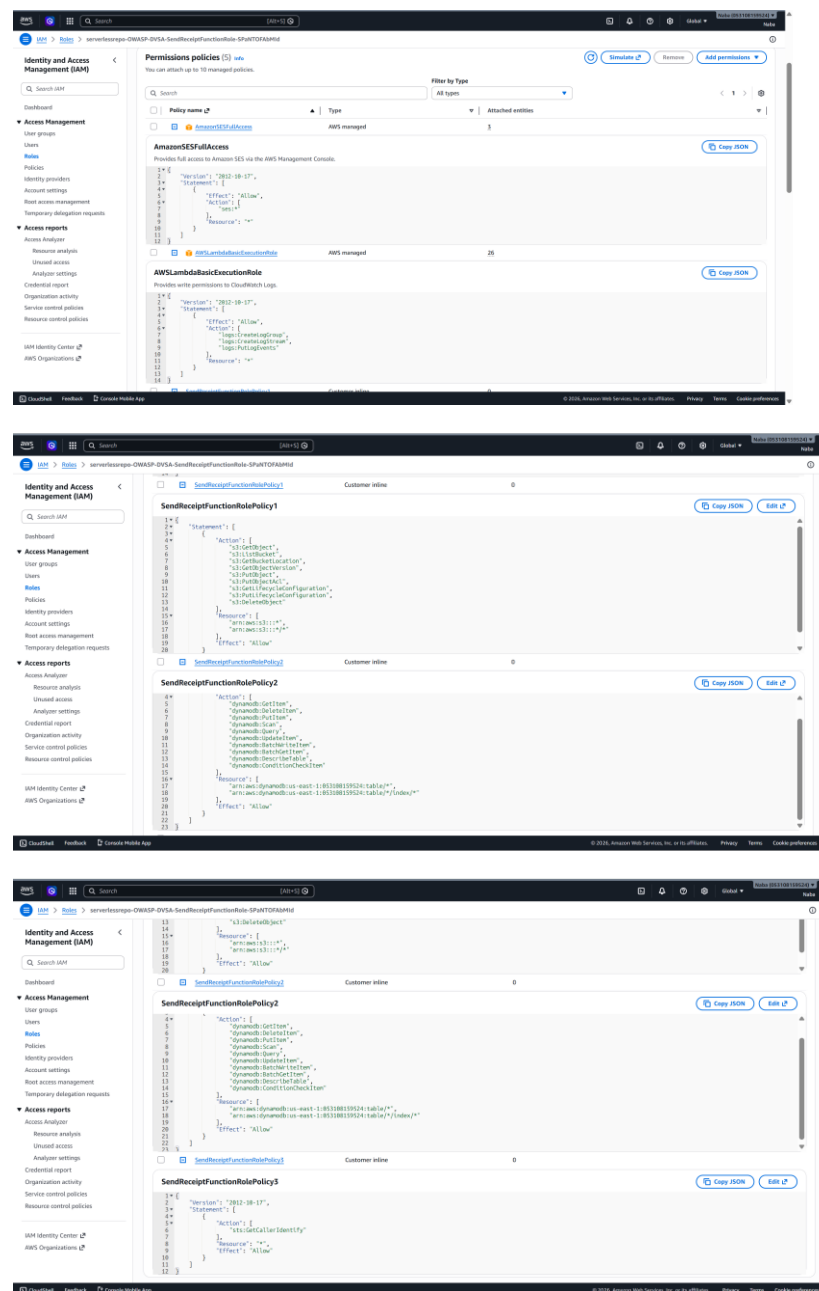
The second screenshot shows a detailed view of a log event, revealing a dictionary of temporary AWS credentials (AWS_SESSION_TOKEN, AWS_ACCESS_KEY_ID, AWS_SECRET_ACCESS_KEY) leaked in the plaintext logs.

The dictionary includes temporary AWS credentials such as:

- AWS_SESSION_TOKEN
- AWS_ACCESS_KEY_ID
- AWS_SECRET_ACCESS_KEY

This demonstrates that the Lambda function leaked its temporary credentials in plaintext logs.

2. IAM Role Evidence



The IAM execution role attached to the Lambda function shows overly broad permissions, including wildcard actions such as: `s3:*`, `dynamodb:*`, `ses:*`

This confirms that the role violates the principle of least privilege.

3. AWS CLI Evidence (Unauthorized Access)

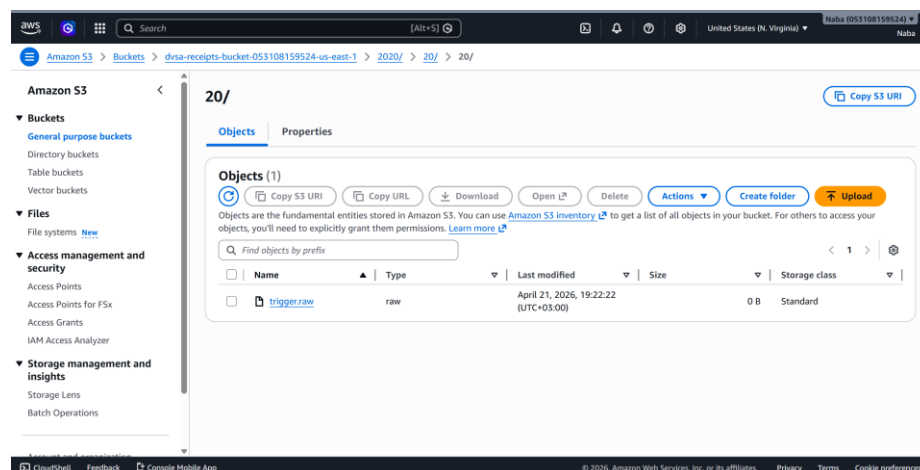
Ubuntu terminal:

```
naba_@LAPTOP-F5TDE2CQ: $ aws dynamodb scan --table-name DVSA-ORDERS-DB --region us-east-1
{
  "Items": [
    {
      "itemList": {
        "M": {
          "1017": {
            "N": "1"
          }
        }
      },
      "totalAmount": {
        "N": "45"
      },
      "orderStatus": {
        "N": "300"
      },
      "confirmationToken": {
```

This proves that the Lambda's temporary credentials can be used to access unrelated AWS resources.

4. S3 Trigger Evidence

S3 bucket path shows the uploaded trigger.raw file under:
2020/20/20/trigger.raw



This confirms that the Lambda was triggered correctly during the test.

Part 6) Fix Strategy / Probable Mitigation

The vulnerability is caused by an over-privileged IAM execution role attached to the DVSA-SEND-RECEIPT-EMAIL Lambda function.

To mitigate the issue, the role must be restricted to the minimum permissions required for the Lambda to perform its intended task.

Fix Strategy

- Remove all wildcard permissions such as `s3:*`, `dynamodb:*`, and `ses:*`.
- Grant only the specific SES permissions needed for sending emails:
 - `ses:SendEmail`
 - `ses:SendRawEmail`
- Keep only the required CloudWatch Logs permissions:
 - `logs:CreateLogGroup`
 - `logs:CreateLogStream`
 - `logs:PutLogEvents`
- Use CloudTrail's "Used permissions" feature to confirm which actions the Lambda actually performs.
- Apply the principle of least privilege to ensure the Lambda cannot access unrelated AWS services.

Part 7) Code / Config Changes

Before (Vulnerable IAM Policies)

1. AmazonSESFullAccess (AWS Managed Policy)

This policy grants full SES access:

AmazonSESFullAccess

Provides full access to Amazon SES via the AWS Management Console.

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Allow",  
6       "Action": [  
7         "ses:*"  
8       ],  
9       "Resource": "*"   
10    }  
11  ]  
12 }
```

This allows the Lambda to send any SES email, manage identities, and perform administrative SES actions.

2. SendReceiptFunctionRolePolicy1 (Inline Policy — S3)

This policy grants broad S3 permissions across all buckets:

SendReceiptFunctionRolePolicy1

```
1 {  
2   "Statement": [  
3     {  
4       "Action": [  
5         "s3:GetObject",  
6         "s3:ListBucket",  
7         "s3:GetBucketLocation",  
8         "s3:GetObjectVersion",  
9         "s3:PutObject",  
10        "s3:PutObjectAcl",  
11        "s3:GetLifecycleConfiguration",  
12        "s3:PutLifecycleConfiguration",  
13        "s3:DeleteObject"  
14      ],  
15      "Resource": [  
16        "arn:aws:s3:::*",  
17        "arn:aws:s3:::*/*"  
18      ],  
19      "Effect": "Allow"  
20    }  
  ]  
}
```

This gives the Lambda the ability to read, write, and delete objects in any S3 bucket.

3. SendReceiptFunctionRolePolicy2 (Inline Policy — DynamoDB)

This policy grants full access to all DynamoDB tables in the account:

SendReceiptFunctionRolePolicy2

```
4   "Action": [  
5     "dynamodb:GetItem",  
6     "dynamodb:DeleteItem",  
7     "dynamodb:PutItem",  
8     "dynamodb:Scan",  
9     "dynamodb:Query",  
10    "dynamodb:UpdateItem",  
11    "dynamodb:BatchWriteItem",  
12    "dynamodb:BatchGetItem",  
13    "dynamodb:DescribeTable",  
14    "dynamodb:ConditionCheckItem"  
15  ],  
16  "Resource": [  
17    "arn:aws:dynamodb:us-east-1:053108159524:table/*",  
18    "arn:aws:dynamodb:us-east-1:053108159524:table/*/index/*"  
19  ],  
20  "Effect": "Allow"  
21 }  
22 ]  
23 }
```

This allows the Lambda to read, modify, and delete data across all DynamoDB tables.

4. SendReceiptFunctionRolePolicy3 (Inline Policy — STS)

This policy allows identity inspection

SendReceiptFunctionRolePolicy3

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Action": [  
6         "sts:GetCallerIdentify"  
7       ],  
8       "Resource": "*",  
9       "Effect": "Allow"  
10    }  
11  ]  
12 }
```

After (Least-Privilege IAM Policy)

To fix the vulnerability, all broad S3, DynamoDB, SES, and STS permissions must be removed.

The Lambda only needs:

- Permission to send emails using SES
- Permission to write logs to CloudWatch

AmazonSESFullAccess_update

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Allow",  
6       "Action": "ses:SendEmail",  
7       "Resource": "*"   
8     }  
9   ]  
10 }
```

Deleted the original AmazonSESFullAccess and add new inline policy.

SendReceiptFunctionRolePolicy1

```
1 {  
2   "Statement": [  
3     {  
4       "Action": [  
5         "s3:GetObject",  
6         "s3:ListBucket",  
7         "s3:GetBucketLocation",  
8         "s3:GetObjectVersion",  
9         "s3:PutObject",  
10        "s3:PutObjectAcl",  
11        "s3:GetLifecycleConfiguration",  
12        "s3:PutLifecycleConfiguration",  
13        "s3:DeleteObject"  
14      ],  
15      "Resource": [  
16        "arn:aws:s3:::dvsa-receipts-bucket-053108159524-us-east-1/*"  
17      ],  
18      "Effect": "Allow"  
19    }  
20  ]  
}
```

SendReceiptFunctionRolePolicy2

```
1  {
2    "Statement": [
3      {
4        "Action": [
5          "dynamodb:GetItem"
6        ],
7        "Resource": [
8          "arn:aws:dynamodb:us-east-1:053108159524:table/DVSA-ORDERS-DB"
9        ],
10       "Effect": "Allow"
11     }
12   ]
13 }
```

Policy3 only allows sts:GetCallerIdentity which is safe, so no need to change it.

Part 8) Verification After Fix

After applying the least-privilege IAM policy, the same exploitation steps were repeated to verify that the vulnerability is no longer exploitable. The expected behavior is that unauthorized operations should now be blocked, while legitimate DVSA functionality continues to work normally.

Amazon S3

168 Action(s) s...

Select All

Deselect All

Reset Contexts

Clear Results

Run Simulation

► Global Settings ⓘ

Action Settings and Results [170 actions selected, 0 actions not simulated, 1 actions allowed, 169 actions denied.]

	Service	Action	Resource Type	Simulation Resource	Permission
►	Amazon SES	SendEmail	identity	*	allowed 1 matching statements.
►	Amazon DynamoDB	Scan	table	*	denied Implicitly denied (no matchin...
►	Amazon S3	AbortMultipartUpload	not required	*	denied Implicitly denied (no matc...
►	Amazon S3	AssociateAccessGrantsIden...	accessgrantsinstance	*	denied Implicitly denied (no matc...
►	Amazon S3	BypassGovernanceRetention	not required	*	denied Implicitly denied (no matc...
►	Amazon S3	CreateAccessGrant	accessgrantslocation	*	denied Implicitly denied (no matc...
►	Amazon S3	CreateAccessGrantsInstance	accessgrantsinstance	*	denied Implicitly denied (no matc...
►	Amazon S3	CreateAccessGrantsLocation	accessgrantsinstance	*	denied Implicitly denied (no matc...
►	Amazon S3	CreateAccessPoint	accesspoint	*	denied Implicitly denied (no matc...
►	Amazon S3	CreateAccessPointForObject...	objectlambdaaccess...	*	denied Implicitly denied (no matc...
►	Amazon S3	CreateBucket	bucket	*	denied Implicitly denied (no matc...
►	Amazon S3	CreateBucketMetadataTable...	bucket	*	denied Implicitly denied (no matc...
►	Amazon S3	CreateJob	not required	*	denied Implicitly denied (no matc...

naba@LAPTOP-F5TDEZCQ: \$ aws dynamodb scan --table-name DVSA-ORDERS-DB --region us-east-1

aws: [ERROR]: An error occurred (AccessDeniedException) when calling the Scan operation: User: arn:aws:iam::053108159524:user/attacker is not authorized to perform: dynamodb:Scan on resource: arn:aws:dynamodb:us-east-1:053108159524:table/DVSA-ORDERS-DB because no identity-based policy allows the dynamodb:Scan action

Part 9) Structured Operation and Security Analysis

Table 1: Structured analysis summary — Table A

Vulnerability	Intended Rule(s)	Artifacts Used to Infer Rule	Normal Behavior Evidence	Exploit Behavior Evidence
Over-Privileged	Lambda must only send SES emails; must not access DynamoDB; temporary credentials must not enable privilege escalation	• CloudWatch logs • AM policies • AWS CLI output • Policy Simulator	Lambda sends SES email only; no DynamoDB access	Successful DynamoDB Scan using leaked credentials

Table 2: Structured analysis summary — Table B

Vulnerability	Why This Is a Deviation	Deviation Class	Fix Applied (Where)	Post-Fix Verification
Over-Privileged	• Violates least-privilege • Expose sensitive data • Allows privilege escalation	Accidental Misconfiguration	IAM Role policy updated to SES-only permissions	• DynamoDB Scan now returns AccessDenied • Policy Simulator confirms restrictions

Part 10) Takeaway / Lessons Learned

In serverless environments, IAM roles define the security boundary, so any misconfiguration directly exposes backend data. The key lesson is that serverless functions must be granted only the exact permissions required, and nothing more.

Applying least-privilege IAM policies and validating them with tools like IAM Policy Simulator prevents similar issues and ensures secure, predictable behavior.